# Peeling Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

**Covers these lab items:** Lye peeling machine

## What it is used for

A lye peeling machine removes the outer skin of suitable fruits or vegetables using a controlled alkaline solution. Correct operation helps loosen the peel while protecting food quality and operator safety.

## Main parts

- Identify the product feeding hopper or loading basket.
- Locate the lye solution tank and designated filling point.
- Identify the heating system and temperature control, if fitted.
- Locate the product conveyor or treatment chamber.
- Identify the rinsing section used to remove residual lye and loosened peel.
- Locate the discharge outlet and waste collection arrangement.
- Identify chemical-resistant seals, protective guards and emergency controls.

## Before you start

1. Read the operating instructions and chemical safety information before use.
2. Operate only under trained supervision with appropriate chemical-handling procedures.
3. Wear chemical-resistant gloves, goggles or face protection, apron and suitable footwear.
4. Inspect tanks, pipes, valves, seals and guards for leaks or damage.
5. Confirm that the lye concentration and temperature follow the approved process instructions.
6. Ensure adequate ventilation and access to an appropriate eyewash and emergency rinse facility.
7. Check that the rinsing system and waste collection arrangements are ready.

## How to use

1. Confirm the machine is clean and the chemical tank is correctly prepared.
2. Load suitable fruits or vegetables into the designated feeding arrangement.
3. Set the approved treatment time, temperature and solution concentration.
4. Start the machine according to the operating instructions.
5. Introduce the product into the treatment chamber using the designated feed system.
6. Monitor treatment conditions and prevent product overexposure to the lye solution.
7. Transfer treated products through the designated rinsing section.
8. Rinse thoroughly using the validated process to remove residual lye and loosened peel.
9. Inspect a sample to confirm acceptable peeling and rinsing results.
10. Stop the machine and isolate power before draining, cleaning or servicing.

## ⚠ Safety

- Lye is corrosive and can cause severe skin and eye burns.
- Never touch the solution or open a chemical tank during operation.
- Never mix lye with acids or other chemicals unless an approved procedure specifically requires it.
- Follow the chemical supplier's instructions for safe preparation and handling; add lye to water slowly, never water to concentrated lye.
- Keep emergency eyewash and rinsing facilities accessible and seek urgent medical assistance after exposure.
- Never release alkaline waste without following approved handling and disposal requirements.

## Care & storage

- Isolate power and follow the approved procedure before draining the solution.
- Clean product-contact surfaces and rinse sections according to the sanitation instructions.
- Inspect tanks, pipes, valves and seals regularly for corrosion or leakage.
- Store chemicals in labelled, closed containers away from food and unauthorised users.

## Common mistakes to avoid

- Using an unverified lye concentration or treatment temperature.
- Failing to rinse treated produce adequately before further processing.
- Handling alkaline solution without appropriate PPE.
- Mixing incompatible chemicals or disposing of lye waste carelessly.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*